DIRECT BROADCAST SATELLITES AND SPACE LAW

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I. INTRODUCTION

In counseling some students in 1886, Mr. Justice Holmes said:

> All that life offers any man from which to start his thinking or his striving is a fact. . . . For every fact leads to every other by the path of the air. Only men do not yet see how, always. And your business as thinkers is to make plainer the way from *some* thing to the *whole* of things; to show the rational connection between your fact and the frame of the universe. . . To be master of any branch of knowledge, you must master those which lie next to it; and thus to know anything you must know all.[1] [Emphasis added]

This observation is particularly relevant to direct broadcast satellites whose technology, when developed, will set the factual parameters within which the pattern of law must be shaped to bring about orderly social progress. Both science and law must start with an evaluation of existing facts and with assumptions concerning the foreseeable future. The revolution in science and technology imposes an obligation on the scientific community to keep the legal profession informed of the latest developments. National and international lawyers have the matching responsibility of absorbing new facts into the framework of social order.

Direct broadcast satellites represent a special development of space communications which is one of the uses of the space environment. From the beginning of the space age, there was recognition that space science and technology produced two reasons for space activities: exploration *and* use, both of which were to be undertaken for the benefit of all mankind. This recognition was embodied in the 1967 Treaty on Outer Space[2] whose provisions evolved over a nine-year period when space technology brought about simultaneously an increase in man's knowledge of the Universe and numerous practical uses for activities on Earth. Data from orbiting spacecraft provided a new technological tool to be used in improving and expanding services required on Earth: swifter communications, more accurate weather prediction, safer navigation, new approaches in medical research,

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[1] O. W. Holmes, The Profession of the Law, in Collected Legal Papers 30 (1937).

[2] Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, Including the Moon and Other Celestial Bodies, January 27, 1967, [1967, pt. 3] 18 U.S.T. 2410, T.I.A.S. No. 6347, 610 U.N.T.S. 205 (effective Oct. 10, 1967) [hereinafter cited as Outer Space Treaty].

increased accuracy of mapping, and many other pursuits. The actuality of using space technology in practical ways led to outstanding international cooperation, particularly on global activities such as communications.

It is evident from the documentary record of the United Nations, particularly its Committee on the Peaceful Uses of Outer Space, that there has been a prevailing and realistic appreciation of the interdependence of technology, institutional requirements, and the legal implications of space programs.[3] Decisions were made on the basis of an amalgamation of these relevant factors, and it was possible to move forward toward a consensus, often expressed in resolutions and draft international agreements.

Among the commonly held basic assumptions was that the 1967 Treaty on Outer Space consisted of general guidelines which could be implemented in a more detailed manner by subsequent international agreements. Building upon, but not contravening the Treaty's provisions, new instruments could be drawn as required by developments in space science and technology. This is the philosophy which underlies the formulation of the three treaty texts worked out since 1967. The 1968 Agreement on the Rescue of Astronauts, the Return of Astronauts and the Return of Objects Launched into Outer Space[4] is the "child" of the "mother" provision in the Outer Space Treaty's article V which prescribes that States shall regard astronauts as envoys of mankind and render them all possible assistance. This Agreement adds necessary details concerning the return of astronauts and space objects and is in harmony with other basic guidelines of the Treaty.

The 1972 Convention on International Liability for Damage Caused by Space Objects[5] is also an harmonious outgrowth of Treaty provisions concerning the responsibility of launching States for space activities. Similarly, the approved 1974 text of the Draft Convention on Registration of Objects Launched into Outer Space stems from concepts of registration and notification to the UN Secretary General that first appeared in the 1967 Treaty.

The line of development of these international space agreements has been consistent in recognizing the Treaty as giving direction through general principles which have been extended into more specific measures as deemed necessary in particular circumstances. Up to the present point in the development of space treaties, it would be possible to codify all provisions into a coherent document whose principles would not be conflicting or internally inconsistent.

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[3]*See generally* U.N. Doc. A/AC.105/66 (1969).

[4]Agreement on the Rescue of Astronauts, the Return of Astronauts, and the Return of Objects Launched into Outer Space, April 22, 1968, [1968, pt. 6] 19 U.S.T. 7570, T.I.A.S. No. 6599, 672 U.N.T.S. 119 (effective Dec. 3, 1969).

[5]Convention on International Liability for Damage Caused by Space Objects, March 29, 1972, [1973, pt. 2] 24 U.S.T. 2389, T.I.A.S. No. 7762 (effective Oct. 9, 1973).

From the time the space age started until the advent of experimental earth resources satellites and the potential of direct broadcast satellites, the development of space science and technology was not hampered by restrictive views of sovereignty. When space programs began, no State objected to the overflight of satellites over its territory.[6] Acceptance was unanimous in producing a consensus that led to article II of the Space Treaty providing that "Outer space, including the moon and other celestial bodies, is not subject to national appropriation by claim of sovereignty, by means of use or occupation, or by any other means."[7] Sovereign airspace and non-sovereign outer space, undefined in their relations, could have become an insuperable obstacle to international agreement, but instead, this issue was set aside and a practicable basis for agreement was substituted, namely, the functional approach that States are responsible for space activities whether their spacecraft are located on the ground, in airspace or in outer space.

In spite of this history of international space development, characterized by a blending of technological, political, economic and legal factors, the advent of the experimental earth resources satellite and the possibility of direct broadcast satellites raised issues of sovereignty which led to proposals for various forms of national control over international space activities.[8] In the case of earth resources satellites, there was fear that pictures taken from outer space would be used by the launching nation to exploit the national resources of other States; in the case of direct broadcast satellites, there was fear that free flow of information would result in receiving nations being subjected to undesirable broadcasts. In both cases, the argument was advanced that the 1967 Treaty on Outer Space did not cover earth resources and direct broadcast satellites because these particular space applications were not known at the time the Treaty was being drafted. Additionally, the claim was made that the Treaty applies only to outer space and not to activities on earth. In analyzing these points of view, this paper will deal only with direct broadcast satellites and will trace the technological developments occurring during the same period of time when the Treaty's provisions were being formulated.

II. THE UNITED NATIONS, OUTER SPACE TREATY AND SPACE COMMUNICATIONS

The text of the Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, Including the Moon and Other Celestial Bodies was adopted unanimously by the UN General Assembly in passing Resolution 2222 (XXI) on December 19, 1966. The concepts incorporated in this treaty text, which was recommended to nations for adoption, began to develop in 1958. On December 13, 1958, General Assembly Resolution 1348 (XIII) emphasized the common interest of mankind

[6]S. Lay & H. Taubenfeld, The Law Relating to Activities of Man in Space 37-62 (1970) [hereinafter cited as S. Lay & H. Taubenfeld]. *See also* E. Galloway, International Cooperation and Organization for Outer Space, S. Doc. No. 56, 89th Cong., 1st Sess. (1965) [hereinafter cited as E. Galloway].

[7]Outer Space Treaty, *supra* note 2, 18 U.S.T. at 2413.

[8]U.N. Doc. A/AC.105/127 (1974) and Annexes I-VI.

in outer space which should be used only for peaceful purposes for the benefit of mankind. There was recognition that outer space activities could increase knowledge and improve life, and vigorous pursuit of programs of international and scientific cooperation was favored.[9]

When General Assembly Resolution 1472 (XIV) created the UN Committee on the Peaceful Uses of Outer Space on December 12, 1959, it was clear that space was not meant merely for exploration but was also to be used for "the betterment of mankind", "the development of science and the improvement of the well-being of peoples."[10]

On September 22, 1960, President Eisenhower addressed the General Assembly, identifying the peaceful uses of outer space as "[b]etter weather forecasting, improved worldwide communications, and more effective exploration not only of outer space but of our own Earth—these are but a few of the benefits of such cooperation."[11]

On December 20, 1961, General Assembly Resolution 1721 (XVI) passed unanimously, providing that outer space should be used only for mankind's betterment and for the benefit of States regardless of the level of their economic or scientific development. The resolution indicated UN awareness of worldwide benefits to be derived from weather research and analysis and recommended the study of measures to "develop existing weather forecasting capabilities and to help Member States make effective use of such capabilities through regional meteorological centres." Furthermore, five years before the Treaty draft was concluded, there was specific recognition that outer space be used for activities conducted on the Earth and not solely for activities confined to outer space itself.

The General Assembly

> Invites the Special Fund and the Expanded Programme of Technical Assistance, in consultation with the International Telecommunication Union, to give sympathetic consideration to requests from Member States for technical and other assistance for the survey of their communication needs and for the development of their domestic communication facilities so that they may make effective use of space communication . . . .[12]

In that same year, on September 25, 1961, President Kennedy addressed the General Assembly and called for extending the rule of law to outer space, saying that "We shall propose, finally, a global system of communications satellites linking the whole world in telegraph and telephone and radio and television."[13]

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[9] 12 Y.B. of the U.N. 22-23 (1958).

[10] 13 Y.B. of the U.N. 28-29 (1959).

[11] 43 Dep't State Bull. 554-55 (1960).

[12] 15 Y.B. of the U.N. 36 (1961).

[13] 45 Dep't. State Bull. 622 (1961).

The following year, on December 19, 1962, the General Assembly adopted unanimously its Resolution 1802 (XVII), again emphasizing the use of outer space as a means for improving earthly conditions. UN agencies were invited "to give sympathetic consideration to requests from Member States for technical and financial assistance to supplement their own resources for these activities, including the improvement of meteorological networks." The General Assembly

> Believes that communication by satellite offers great benefits to mankind as it will permit the expansion of radio, telephone and television transmissions, including the broadcast of United Nations activities, thus facilitating contact among the peoples of the world.[14]

The role of the International Telecommunication Union in the allocation of radio frequencies was dealt with in specific terms.

On December 13, 1963, the General Assembly passed unanimously two basic documents which were direct forerunners of the Treaty on Outer Space which was to be concluded three years later: Resolution 1962 (XVIII) on Legal Principles and Resolution 1963 (XVIII) on International Cooperation. In the latter resolution, the use of outer space for functions to be performed on the Earth was implicit in the paragraph noting "that the United States and the Soviet Union have reached an agreement looking toward cooperation in the fields of satellite meteorology, communications and magnetic field mapping." The General Assembly also welcomed decisions of the International Telecommunication Union on "the allocation of frequency bands for space communication and procedures for their use as a step in the development of space radio communications." A progress report was requested and the connection between outer space and the Earth was apparent in the resolution which

> Recognizes the potential contribution of communications satellites in the expansion of global telecommunications facilities and the possibilities this offers for increasing the flow of information and for furthering the objectives of the United Nations and its agencies.[15]

In 1964, the UN Committee on the Peaceful Uses of Outer Space reported that it had "decided to consider questions relating to the use of satellites for transmitting radio and television programmes intended for direct reception by the general public after the report of the International Radio Consultative Committee (CCIR) on this subject had been received by ITU." Also noted was the traffic control for sea and air, a development which might lead to a global civil navigation satellite system.[16]

On December 21, 1964, the General Assembly adopted unanimously Resolution 2130 (II and III) (XX) welcoming the progress reports of the International Telecommunication Union (ITU).

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[14] 16 Y.B. of the U.N. 54 (1962).

[15] 17 Y.B. of the U.N. 109 (1963).

[16] 18 Y.B. of the U.N. 83 (1964).

In 1965, the Scientific and Technical Subcommittee of the Committee on the Peaceful Uses of Outer Space considered the fourth progress report of the ITU:

> These studies dealt with radio communications for satellites and research space probes, such as signal relay, *direct broadcasting,* meteorology, and navigational aids. Questions relating to radio astronomy, sharing of radio-frequency bands, multiple access to communication satellite systems and the problems of telegraph and telephone techniques had also been examined during the period under review by various ITU bodies.[17] [Emphasis added]

It is evident from the above 1965 report, which refers to earlier studies, that direct broadcast satellites had been identified and analyzed within the United Nations some years before the draft text of the Treaty on Outer Space was adopted by the General Assembly on December 19, 1966. This is not surprising when it is considered that 21 years earlier, and 12 years before an artificial earth satellite was orbited, Arthur C. Clarke published an article describing communications from outer space to the Earth and commented that "[a] true broadcast service, giving constant field strength at all times over the whole globe would be invaluable, not to say indispensable, in a world society."[18] Using outer space for activities on Earth was highlighted again in 1966 by UN Resolution 2221, (XXI), also adopted on December 19. This resolution called for a United Nations Conference on the Exploration and Peaceful Uses of Outer Space, the objectives being "to examine the practical benefits of space programmes on the basis of scientific and technical achievements, and the opportunities available to non-space powers for international cooperation in space activities, with special reference to the needs of the developing countries . . . ."[19]

The conference on Space Exploration and Applications was held in Vienna from August 14-27, 1968, and the Proceedings indicate not only a wide variety of uses for space data on Earth, but a definite interest in direct broadcast satellites.[20] By that time the joint project between the United States and India for direct television broadcasting to community receivers was being worked out. There was an explanation of the difference between this type of broadcasting, which does not arouse much controversy, and that potentially attainable by means of augmented and unaugmented broadcasts into home receivers. Television broadcasting directly from a satellite into homes is considered to have more impact on people and it is this particular aspect of space communications which leads to demands for national controls and consequent restrictions on broadcasting states and non-governmental organizations. The technological facts and legal implications were well known, particularly during the final stages when a consensus was forming for adoption of the Treaty on Outer Space.

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[17] 19 Y.B. of the U.N. 33 (1965).

[18] Clarke, Extraterrestrial Relays, in Wireless World (1945), reprinted in A. Clarke, Voices From the Sky: Previews of the Coming Space Age 233-41 (1965).

[19] 20 Y.B. of the U.N. 47-48 (1966).

[20] Proceedings of the United Nations Conference on the Exploration and Peaceful Uses of Outer Space 93-275 (1968).

The continuous interest of the General Assembly was shown when, recalling its 1961 resolution favoring space communications, there was unanimous adoption of Resolution 2453 (XXIII) on December 20, 1968 approving the creation of the Working Group on Direct Broadcast Satellites:

> Approves the establishment by the Committee on the Peaceful Uses of Outer Space of a working group to study and report on the technical feasibility of communication by direct broadcast from satellites and the current and foreseeable developments in this field, including comparative user costs and other economic considerations, as well as the implications of such developments in the social, cultural, legal and other areas, and expresses the hope that interested States Members of the United Nations and members of the specialized agencies will contribute comments and working papers to the working group for its information and guidance to the performance of its task . . . .[21]

III. DEVELOPMENTS IN SPACE COMMUNICATIONS TECHNOLOGY AND ORGANIZATION

I. Technology. All satellites involve communication with the Earth and in this general sense there was awareness of space communications from the time the first satellite was orbited in 1957. The application of space technology to existing communications systems developed so rapidly, however, that it became one of the major ways in which outer space was used on Earth for the benefit of mankind. By 1962 the U.S. Telstar I was broadcasting the first intercontinental television demonstration between Europe and the United States. Among the programs was a tribute to Dag Hammarskjold, Astronaut Schirra's launch from Cape Canaveral, and President Kennedy's address to the nation.[22]

In 1963, Relay I broadcast the death of Pope John XXIII, the coronation of Pope Paul VI, the opening of the UN General Assembly, and the Inaugural television transmission to Japan.

In 1964, Telstar II televised the winter Olympic games from Austria and a press program with the French foreign minister. Relay I televised N. Khrushchev's visit to Hungary, and Relay II broadcast the visit of President de Gaulle to Mexico and later broadcast coverage of the British elections. Telstar II brought TV coverage of President Johnson's inauguration and Churchill's funeral in 1965.[23] During 1966 when the United Nations was completing work on the text of the Treaty on Outer Space, the Applications Technology Satellite (ATS) was launched by NASA on December 6 for experiments in communications.

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[21] 22 Y.B. of the U.N. 62 (1968).

[22] E. Galloway, *supra* note 6, at 58.

[23] *Id.* at 56-59.

2. **Organization.** The rapid development of space communications led the United States to enact the Communications Satellite Act of 1962[24] providing for "a commercial communications satellite system, as part of an improved global communications networks . . . [to] serve the communication needs of the United States and other countries, and which will contribute to world peace and understanding."[25] Global coverage was to be made available as promptly as possible while "care and attention will be directed toward providing such services to economically less developed countries and areas as well as those more highly developed . . . ."[26] Upon the basis of this policy, the Communications Satellite Corporation (COMSAT) was created.

By August 1964, the International Telecommunications Satellite Consortium (INTELSAT) was established in accordance with two interim agreements providing for a global commercial communications satellite system.[27]

Such national and international arrangements were not concerned with direct broadcast satellites which have been the subject of special study and analysis by the United Nations, but it is important to understand the total situation—technological, institutional, economic, and legal—which was developing at almost unprecedented speed in relating space technology to beneficial uses on Earth. Recognition of this development motivated the formulation of guiding legal principles designed to cover all space activities, including communications. There was no assumption that treaty provisions were inapplicable to human activities on Earth just because data was being received from outer space.

IV. THE UNITED STATES POSITION

When the UN Working Group on Direct Broadcast Satellites met in Geneva from March 11-22, 1974, the U.S. Representative, Lee T. Stull, explained that although the United States has "serious reservations about the advisability of adopting binding principles governing such broadcasting,"[28] it recognized that many members favored guidelines. U.S. reservations stem from questions concerning national and international lack of experience with this still undeveloped aspect of space communications, concern lest future technological development be inhibited, and possible adverse effects on the free exchange of ideas and information.[29] Even though the UN estimated that direct televi-

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[24]Act of Aug. 31, 1962, Pub. L. No. 87-624, 76 Stat. 419.

[25]*Id.*

[26]*Id.*

[27]E. Galloway, *supra* note 6, at 50-54, 542-53. *See also* S. Lay & H. Taubenfeld, *supra* note 6, at 119-35.

[28]*See* Hearings on S. 2955 Before the Senate Comm. on Aeronautical and Space Sciences, 93rd Cong., 2nd Sess., pt. 2 at 772 (1974).

[29]*See Id.*

sion satellite broadcasts into individual home receivers is unlikely before 1985, and there are at present no U.S. plans for development, nevertheless, the United States decided that a valuable contribution could be made by drafting voluntary principles based upon ideas on which there appeared to be general agreement at past meetings. To assist in identifying common interests which might constitute a basis for international agreement, the U.S. Working Paper of Draft Principles on Direct Broadcast Satellites was presented on March 11, 1974.[30]

The *first* of the eleven U.S. draft principles provided that international direct television broadcasting by satellites should be conducted in accordance with international law, the United Nations Charter, the Treaty on Outer Space, the Universal Declaration of Human Rights and the Declaration on Principles of International Law Concerning Friendly Relations and Cooperation Among States.

Second, there should be compliance with the technical parameters and procedures of the ITU Convention and Radio Regulations.

Third, international peace and security, cooperation and friendly relations should be maintained.

Fourth, free and open exchange of ideas and information should be encouraged and expanded, differences among cultures should be considered, and there should be maximum use of benefits.

Fifth, every State is entitled to send and receive broadcasts and to share in benefits.

Sixth, States and international organizations should cooperate to assist other nations, particularly developing countries. This can be accomplished by training in technology, program production, program and personnel exchanges, and by considering the creation of regional centers.

Seventh, States with similar interests should consider regional and international methods.

Eighth, international professional contributions should be encouraged.

Ninth, international cooperation among broadcasters and regional broadcast associations should be recognized as desirable and be promoted.

Tenth, disagreements should be settled through consultation and established procedures.

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[30]U.N. Doc. A/AC.105/WG.3(V) CRP. 2 (1974). *See also* Statement by U.S. Representative at the Fifth Session of the U.N. Working Group on Direct Broadcast Satellites, Press Release at Geneva, Switzerland, March 13, 1974. Statement reprinted in Hearings on S. 2955 Before the Senate Comm. on Aeronautical and Space Sciences, 93rd Cong., 2nd Sess., pt. 2 at 776-78 (1974).

Eleventh, as practical experience warrants, questions concerning international direct television broadcasting should be reviewed by the United Nations and member States.

The U.S. draft principles do not include a provision that consent must be obtained by a broadcasting state from receiving states prior to international direct television broadcasting by satellites. Prior consent, the major issue in official discussions, is decisively rejected by the United States, major reasons being opposition to government censorship of program content and interference with the free flow of information and ideas. The proposal of the Soviet Union, for example, specifies that "States may carry out direct television broadcasting by means of artificial earth satellites to foreign States only with the express consent of the latter," and such broadcasts "may be carried out only by organizations which are under the control of governments of the States concerned." Furthermore, advertising and commercial broadcasts would require specific agreement.[31] Views similar to those of the USSR have been expressed by other delegations.[32]

The U.S. Representative to the Working Group therefore asked Members to discuss questions pertinent to prior consent: (1) How can the absolute veto by a State over TV broadcasts be reconciled with article 19 of the Universal Declaration of Human Rights which contains the right to "receive and impart information and ideas through any media and regardless of frontiers?" (2) What effect would selective censorship have on expanding the use of satellite broadcasting? (3) What would be the effects of arbitrary withdrawal of consent, particularly on regional areas? (4) What are the implications for domestic systems of programs that spill over the boundaries of other nations? How can practical distinctions be made between intentional spillover designed to affect public opinion and unintentional spillover which unavoidably results from the configuration of this technology? (5) What effect would the prior consent principle have on other communications media? Would it be a precedent resulting in inhibiting the free flow of information and ideas? Finally, the U.S. Representative explained that objections to prior consent did not mean that the activity would be abused if the technology were ever developed.[33]

It was clear from the discussions of various delegations to the Working Group that there was no agreement on a definition of "prior consent." To some nations it meant prior consent to launching an operational system; to others it meant a general overview of broadcasting; and to still others, prior consent meant government censorship of the content of each individual program. The principal element underlying the United States position is a desire to stress the potential benefits and to facilitate realization of the benefits flowing from direct broadcast satellite technology.

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[31]U.N. Doc. A/AC.105/127 (1974), Annex II at 1-4.

[32]*Id.*, Annex III at 1-4 (Canada and Sweden); Annex V at 1-7 (Argentina).

[33]Statement by U.S. Representative, *supra* note 30. Also printed in Hearings, *supra* note 30, at 775-76.

V. STATUS OF UNITED NATIONS CONSIDERATION OF DIRECT TELEVISION BROADCASTING FROM SATELLITES

In May 1976 the Legal Subcommittee of the UN Committee on the Peaceful Uses of Outer Space is scheduled to meet in Geneva and continue its consideration of direct television broadcasting from satellites. At that time it will have before it not only the 1974 report of the Working Group on Direct Broadcast Satellites[34] but also the report of the Legal Subcommittee resulting from its meetings at UN Headquarters in New York from February 10 to March 7, 1975.[35] The subject was handled by the Legal Subcommittee as a matter of high priority, fourteen component parts or issues were identified, and areas of agreement and disagreement were delineated. The 1976 meeting will discuss the following subjects to determine whether or not a consensus can emerge.

1. **Purposes and Objectives.** Two alternatives were advanced, each including as purposes of direct television broadcasting by satellite the maintenance of international peace and security, enhancing the educational level worldwide and assisting developing countries. Alternative A, however, would require States to carry out such activities "exclusively" in a manner compatible with all stated objectives and with due regard to a principle on which a consensus has not yet been reached: the applicability of international law. Alternative B is not expressed in mandatory terms, preferring that activities should facilitate, promote and encourage beneficial applications. The United States position is that direct television broadcasting from satellites concern "international" activities.

2. **Applicability of International Law.** Although there is a consensus on including the United Nations Charter, the 1967 Outer Space Treaty, the International Telecommunication Union Convention and its Radio Regulations in the international law applicable to this subject, no consensus has been reached on inclusion of the generally recognized rules of international law, the Declaration on Principles of International Law concerning Friendly Relations and Cooperation among States, the Universal Declaration of Human Rights, and the International Covenant on Civil and Political Rights. Agreement is also lacking on whether or not the entire principle would be mandatory.

3. **Rights and Benefits [of States].** This is one of the most thorny issues because it evokes basic disagreement on whether States or individuals have a right to enjoy the benefits of direct television broadcasting from satellites and whether States should supervise such activities.

4. **International Cooperation.** Although there is agreement on international cooperation with States or their authorized entities making appropriate arrangements, positions differ on whether activities "shall" or "should" be undertaken.

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[34]U.N. Doc. A/AC.105/127 (1974) and Annexes I-VI. *See also* earlier reports: U.N. Doc. A/AC.105/51 (1969); A/AC.105/66 (1969); A/AC.105/83 (1970); A/AC.105/117 (1973).

[35]U.N. Doc. A/AC.105/147 (1975) and Annex II. *See also* U.N. Doc. A/AC.105/133 (1974) and Annexes I-V.

5. **State Responsibility.** A consensus has been reached on this principle which is carried over from the 1967 Outer Space Treaty to provide for compliance by States and international organizations specifically with reference to direct television broadcasting by satellites.

6. **Consent and Participation.** Differences of opinion on this basic issue created two incompatible alternatives. Alternative A requires the consent of a State toward which direct television broadcasts from satellites are beamed, and the consenting State would also have the right to participate in such activities. Consent and participation would not be required, however, if the broadcasts were technically unavoidable under regulations of the International Telecommunication Union. Alternative B does not require prior consent but provides that the broadcasting State should consult with a receiving State concerning any restrictions that might be imposed. Compatibility with international law is stressed, particularly as related to freedom of expression "which includes freedom to seek, receive and impart information and ideas of all kinds, regardless of frontiers." This alternative would recognize technical restrictions imposed by international law.

7. **Spill-Over.** The entire concept of spillover is in square brackets, thus indicating disagreement on its provisions. Alternative A would require the maximum use of technical means by the broadcasting State to reduce radiation over other countries unless prior agreement with them had been reached. The Alternative B proposal, without prejudice to regulations of the International Telecommunication Union, provides that all reasonable means would be used to reduce unintended radiation to a minimum.

8. **Program Content.** Wide disagreement is expressed by the three bracketed paragraphs on the content of programs. Some nations call for cooperation on program content, production and interchange of programs; others want agreements on broadcasting commercial advertising; others wish to exclude programs which undermine "the foundations of the local civilization, culture, way of life, traditions or language." Some nations, preferring the free flow of information and ideas, would be against restrictive provisions.

9. **Unlawful/Inadmissible Broadcasts.** The concept of unlawful broadcasts which certain States regard as inadmissible because of undesired program content is antithetical to the free flow of information and ideas favored by other States, and consequently there is sharp disagreement on this point.

10. **Duty and Right to Consult.** Two alternatives reflect different viewpoints on reasons for consultation between States. Alternative A requires a State to consult without delay if another considers it will be adversely affected by direct television broadcasts from satellites. Alternative B provides that any matter of mutual concern can be the occasion for consultations held without delay.

11. **Peaceful Settlement of Disputes.** A consensus was reached on the desirability of settling disputes by consultations or other established procedures.

12. **Copyright, Neighboring Rights and Protection of Television Signals.** Agreement was reached on bilateral and multilateral protection of such rights and giving special attention to developing countries, but differences remained on the general statement of rights not being affected by such broadcasts as well as on the idea that the Brussels 1974 Convention relating to program distribution should not be affected.

13. **Notification to the United Nations System.** Agreement on informing the UN Secretary General and his distribution of information followed the guideline in article XI of the 1967 Outer Space Treaty, and the only disagreement was on "including information on the contents of programs."

14. **Disruption.** Agreement was reached on preventing disruption to communications concerned with the safety of life.

VI. CONCLUDING OBSERVATIONS

Direct broadcast satellites constitute a complicated problem because so many interrelated factors are involved: technological, political, economic, cultural, and legal. Although numerous points of view have been expressed concerning the mitigation or solution of perceived problems, some progress has resulted from indepth studies made by the United Nations and its relevant specialized agencies, individual nations, and other governmental and nongovernmental organizations. This progress consists of identification of different parts of the total problems, presentation of issues, and some realization of first steps toward probable areas of agreement. Sufficient momentum in official discussions of the subject has been attained to accomplish inclusion of Direct Broadcast Satellites on the agenda of the Committee on the Peaceful Uses of Outer Space as a "priority item."

The technological features of Direct Broadcast Satellites have been fully set forth at UN meetings, but they are not understood by all discussants, particularly regarding the interrelationship between technical and legal matters. In brief, the technology is divided into two main parts: direct broadcasting into community receivers and into individual home receivers. Individual reception is further divided into two parts: broadcasting into augmented home receivers and later into unaugmented home receivers. There is little controversy over such broadcasting to community receivers, a technology which is already being tested in the U.S. ATS–6 satellite. There is an agreement between the United States and India for direct satellite broadcasting of Indian programs into their community receivers.

The controversial aspects of the question concern the possibility of direct television satellite broadcasting into home receivers, whether augmented or unaugmented. The fact that this aspect of the technology is not yet developed does not lessen concern by some nations who fear receiving unwanted broadcasts and desire to formulate guidelines for control of the activity prior to its development.

The main issue is whether or not prior consent would have to be obtained from States before programs could be broadcast. Most discussion has been in general terms about the principle of prior consent and its conflict or accommodation with sovereign rights and the free flow of information and ideas. The term has not been defined and there are underlying assumptions concerning its meaning, ranging all the way from consent to launching an operational system to censorship of each individual program.

Sovereignty did not become a controversial issue at the beginning of the space age when it was discussed in somewhat geographical terms of where airspace ends and outer space begins. Technological and legal difficulties in this approach were set aside and a decision incorporated in the Treaty on Outer Space that outer space is free and not subject to sovereign claims. Agreement was reached on the basis of guiding principles for functions to be performed in outer space. General provisions in the Treaty on Outer Space were then extended when specific circumstances arose which required more detailed arrangements, as in the Agreement on Assistance to Astronauts and the Liability Convention. Relevant UN bodies have accepted the applicability of the 1967 Treaty to this subject.

At the present time, study is at the stage where agreement could be sought on (1) those provisions in the Treaty on Outer Space which apply to direct television broadcast satellites, and (2) those included in other existing international law. It would then be possible to arrive at the third stage: decisions, reached by consensus, on any additional matters which might not be covered.